

in complexity and hence is useful for different academic levels."

"The lowest level helps understand the basic concepts of mechanics, working with motors, gears, pulleys and other assemblies. The next level makes use of servo actuators, sensors, control modules and small wheel robots, etc. The higher level deals with flexible hardware, software toolkits for kinematics/coordinated control, machine vision and artificial intelligence algorithms," he adds.

Indian vs imported kits

There are both made-in-India kits as well as imported kits available in the Indian market. You can order a kit even online after matching your requirements with the offered kit's specifications.

"Indian robotic kits are generally targeted at specific applications. Imported kits are generic and expandable. Kits like Lego Mindstorms and TETRIX are very high-quality, flexible and expandable," informs Sudhanshu.

"India was introduced much later to the field of robotics as compared to the developed nations. Even though we are far behind the level that has already been achieved, we are progressing at a fast pace in order to match their level within the next few years," shares Pradeep Kumar Sharma, director, Robosapiens Technologies.

"The Indian robotics kits are in a nascent stage as compared to imported ones. There is lot to be done in finishing and accuracy," Ashutosh Agarwal, managing director, Tesca Technologies, says.

However, Malewar differs: "Indian robotics kits have come a long way since last four years. These are now at par with any international platform."

The price of a robotic kit increases with the level of advancement. Indian robotic kits are thus generally lower priced than imported kits.

Siddharth Dev, CTO & founder, Technido, says, "The price-to-feature ratio is reasonably good for Indian kits but aesthetically they are not as attractive as imported ones. Due to

the low scale of production and sales, most manufacturers make aluminium or other metal chassis that have limitations of bending and producing them into attractive forms."

What's available in the market?

"There are large varieties of robotic kits available in the market, from small two-wheeled basic robots, intermediate multi-wheeled robots to complex robots such as humanoids and hexapods," says Malewar.

"Robotic kits are available from different manufacturers. The difference lies in the functionality and quality. There are many robotic kit manufacturers who offer robots that look just like Lego Mindstorms but have reduced functionality and quality. Some robotic kits are sold as programmable bi-pedal robots. But these really have just three to four modes of operation. This defeats the purpose of buying a robotic kit for educational purpose," cautions Deshpande.

Automation and customisation

There are basically three types of automated robotic kits available:

1. Fully-automatic wherein the system requires only one command and the entire operation is processed under that command

2. Semi-automatic wherein the user has to offer commands to the system after certain levels

3. Manually-controlled wherein the user has to manually operate the system

The various levels of automation and customisation possible in a robotic kit are specific to the type of robotic kit. Programming facility in a robotic kit is of prime importance. The type of processor used, ability to use a processor of choice, ability to add third-party sensors and accessories enhance the usability and expandability of the kit.

"Any good robotics kit should be highly customisable and expandable. All advanced kits should be expandable with custom-built sensors and actuators. Automation should be completely possible through programming

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